Quadratic Equation-1

Quadratic Equation: $ax^2+bx+c=0 \ (a\neq 0)$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

where, $D = b^2-4ac$ (Discriminant)

Quadratic eqⁿ will always have 2 roots:
- $D<0 \Rightarrow$ Imaginary 2 roots
- $D=0 \Rightarrow$ Equal & real 2 roots
- $D>0 \Rightarrow$ Distinct & real 2 roots

Vieta's Formula:

$ax^2+bx+c=0 \rightarrow \alpha, \beta$

* $\alpha+\beta = \frac{-b}{a}$
* $\alpha\beta = \frac{c}{a}$
* $\alpha-\beta = \frac{\sqrt{D}}{|a|}$

Some Important Transformations:→

1) $\alpha^2+\beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$
2) $\alpha^3+\beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$
3) $(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$
5) $\alpha^4+\beta^4 = \left[(\alpha+\beta)^2-2\alpha\beta\right]^2 - 2(\alpha\beta)^2$

Ques If α and β be two roots of the eqⁿ $x^2-64x+256=0$. Then find $\left(\frac{\alpha^3}{\beta^5}\right)^{\frac{1}{8}} + \left(\frac{\beta^3}{\alpha^5}\right)^{\frac{1}{8}}$

Sol'n $S = \frac{\alpha^{3/8}}{\beta^{5/8}} + \frac{\beta^{3/8}}{\alpha^{5/8}} \Rightarrow \frac{\alpha+\beta}{(\alpha\beta)^{5/8}}$

$S = \frac{(64/1)}{(256)^{5/8}} \Rightarrow \frac{64}{(2^8)^{5/8}} \Rightarrow \frac{64}{2^5}$

$S = \frac{2^6}{2^5} \Rightarrow 2$ Ans

Ques If α & β are roots of the eqⁿ $2x(2x+1)=1$, then find β in terms of α.

$2\alpha+2\beta=-1$

$\alpha+\beta=-1/2$

$\alpha\beta=-1/4$

Sol'n $4x^2+2x-1=0 \rightarrow \alpha, \beta$

Since α is a root of given eqⁿ

$4\alpha^2+2\alpha-1=0$

$4\alpha^2-1-2\beta-1=0 \Rightarrow 2\beta = 4\alpha^2-2$

$\boxed{\beta = 2\alpha^2-1}$ Ans

Ques Find product of roots of $9x^2-18|x|+5=0$.

Sol'n Concept: $|x|^2 = x^2$

$9|x|^2-18|x|+5=0$

$9|x|^2-15|x|-3|x|+5=0$

$3|x|(3|x|-5)-1(3|x|-5)=0$

$\Rightarrow (3|x|-5)(3|x|-1)=0$

$|x| = 5/3 \Rightarrow x = \pm 5/3$

$|x| = 1/3 \Rightarrow x = \pm 1/3$

Product of roots $= \frac{1}{3}\times\frac{-1}{3}\times\frac{5}{3}\times\frac{-5}{3} = \frac{25}{81}$

Ans

Ques a, b are distinct real roots of $x^2+20x-2020=0$ and c, d are distinct complex roots of $x^2+20x+2020=0$, then find $ac(a-c)+ad(a-d)+bc(b-c)+bd(b-d)$.

Sol'n $x^2+20x-2020=0 \rightarrow a, b$: $a+b=-20$, $ab=-2020$

$x^2+20x+2020=0 \rightarrow c, d$: $c+d=20$, $cd=2020$

Now,

$S = a^2c-a^2c^2-ad^2+a^2d+b^2c-bc^2+b^2d-bd^2$

$S = a^2(c+d)+b^2(c+d)-c^2(ab)-d^2(ab)$

$S = a^2(20)+b^2(20)-c^2(-20)-d^2(-20)$

$S = 20\left[a^2+b^2+c^2+d^2\right]$

$S = 20\left[(a+b)^2-2ab+(c+d)^2-2cd\right]$

$S = 20\left[(-20)^2-2(-2020)+(20)^2-2(2020)\right]$

$S = 20[800] \Rightarrow 16000$ Ans

Ques $ax^2-x+a=0$ has two distinct real roots x_1 and x_2 satisfying $|x_1-x_2|<1$. Find interval in which a lie.

Sol'n $a^*x^2-x+a=0 \rightarrow x_1, x_2$

$|x_1-x_2|<1$

$\frac{\sqrt{D}}{|a|}<1$

$\sqrt{1-4a^2}<|a|$

$1-4a^2<a^2$

$1-5a^2<0$

$5a^2-1>0$

$\left(a^2-\frac{1}{5}\right)>0$

$\Rightarrow \left(a-\frac{1}{\sqrt{5}}\right)\left(a+\frac{1}{\sqrt{5}}\right)>0$ — ①

Roots are real & distinct

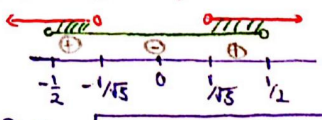
For $D>0$

$1-4a^2>0$

$4a^2-1<0$

$a^2-\frac{1}{4}<0$

$\left(a-\frac{1}{2}\right)\left(a+\frac{1}{2}\right)<0$ — ②



① ∩ ② $\Rightarrow$ $\boxed{a \in \left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right) \cup \left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)}$ Ans

Formation of Quadratic Eqⁿ by Roots:

$x^2 - x(\text{Sum of Roots}) + (\text{Product of roots}) = 0$

$\boxed{x^2 - x(\alpha+\beta) + (\alpha\beta) = 0}$

Newton's Method: (Power of Roots)

Suppose, we have a quad. $ax^2+bx+c=0 \rightarrow \alpha, \beta$

Let, $S_n = \alpha^n \pm \beta^n \ (n>1)$

Then, if we replace,

$x^2 \rightarrow S_n$

$x \rightarrow S_{n-1}$

$1 \rightarrow S_{n-2}$

$\Rightarrow \boxed{a(S_n)+b(S_{n-1})+c(S_{n-2})=0}$

→ Always holds true for $(n>1)$.

Ques α and β are roots of $x^2-6x-2=0$, with $\alpha>\beta$. If $a_n = \alpha^n-\beta^n$, for $n\geq 1$, find $\frac{a_{10}-2a_8}{2a_9}$.

Sol'n $x^2-6x-2=0$

$a_{10}-6a_9-2a_8=0$

$a_{10}-2a_8 = 6a_9 \Rightarrow \boxed{\frac{a_{10}-2a_8}{2a_9} = 3}$ Ans

Ques Let p and q be integers and let α, β be the roots of the eqⁿ $x^2-x-1=0$, where $\alpha\neq\beta$. Let $a_n = p\alpha^n + q\beta^n$, for $n\geq 1$, and $a_4 = 28$. Find value of $p+2q$.

Sol'n $x^2-x-1=0$

$a_4-a_3-a_2=0$

$a_4 = a_3-a_2 \Rightarrow (a_2+a_1)+(a_1+a_0)$

$a_4 = a_1+a_0+a_1+a_1+a_0$

$a_4 = 3a_1+2a_0 \Rightarrow 3(p\alpha+q\beta)+2(p+q)$

$\therefore x^2-x-1=0 \rightarrow \alpha, \beta$: $x = \frac{1\pm\sqrt{5}}{2}$

So,

$28 = 3p\left(\frac{1+\sqrt{5}}{2}\right) + 3q\left(\frac{1-\sqrt{5}}{2}\right) + 2p+2q$

Concept: If rational and irrational terms are in multiplication, then

$a+\sqrt{5}b=0 \Rightarrow a=b=0$

$a+\sqrt{5}b=k \Rightarrow a=k \ \& \ b=0$

$28 = \left(\frac{3p}{2}+\frac{3q}{2}+2p+2q\right) + \left(\frac{3p}{2}-\frac{3q}{2}\right)\sqrt{5}$

$\frac{3p}{2} = \frac{3q}{2} \Rightarrow \boxed{p=q}$

$\frac{7p}{2}+\frac{7q}{2} = 28$

$[\because p=q] \quad 7p = 28 \Rightarrow \boxed{p=q=4}$

Thus, $p+2q = 4+2(4)$

$\Rightarrow \boxed{p+2q=12}$ Ans

Identity:

↳ A quadratic having more than two distinct roots, becomes an Identity.

$ax^2+bx+c=0$

↳ Is an identity if $a=b=c=0$

↓

All three should be zero simultaneously (Intersection wala hai)

Eg: For what value of p, the eqⁿ $(p+2)(p-1)x^2+(p-1)(2p+1)x+p^2-1=0$ has more than two roots?

Sol'n $(p+2)(p-1)=0$

$(p-1)(2p+1)=0$

$(p+1)(p-1)=0$

$\Rightarrow \boxed{p=1}$ Ans

Nature of Roots:

(i) If $D>0 \Rightarrow$ Roots are real & distinct

(ii) If $D=0 \Rightarrow$ Roots are equal

(iii) If $D<0 \Rightarrow$ Roots are imaginary

Important Points :→

1) If Coefficients of quad. are rational ⇒ Then Irrational roots would always occur in pair
$(p+\sqrt{q})$ $(p-\sqrt{q})$

2) If Coefficients of quad. are Real ⇒ Then Imaginary roots always occur in pair
$(p+iq)$ $(p-iq)$

3) Roots of Quadratic would be Rational If
↳ Coefficients are Real
↳ D = (Perfect Square)

Ques Find the quadratic eqn with rational coeff. whose root is $2+\sqrt{3}$.

Sol'n Since Rational Coefficients ⇒ Irrational roots must be in pair

i.e. If $(\alpha = 2+\sqrt{3} \Rightarrow \beta = 2-\sqrt{3})$

So, $\alpha+\beta = 4$, $\alpha-\beta = 1$

Quadratic : $\boxed{x^2 - 4x + 1 = 0}$ Ans

Ques No. of all +ve integral values of α for which roots of quad. $6x^2 - 11x + \alpha = 0$ are rational numbers is ____.

Sol'n D = perfect square
$= (b)^2 - 4(6)(\alpha)$
$D = 121 - 24\alpha$

$\alpha = 1 \longrightarrow \alpha = 97$ ✗
$\alpha = 2 \longrightarrow \alpha = 73$ ✗
$\alpha = 3 \longrightarrow \alpha = 49$ ✓
$\alpha = 4 \longrightarrow \alpha = 25$ ✓
$\alpha = 5 \longrightarrow \alpha = 1$ ✓

$\Rightarrow (\alpha = 3, 4, 5)$ ⇓ 3 integral +ve values.
Ans

Graph of Quadratic Eqn :

* Graph of quadratic is always a parabola.

$a>0$ (Upward Parabola) $a<0$ (Downward Parabola)

* Vertex $(V) = \left(\frac{-b}{2a}, \frac{-D}{4a}\right)$ * y-intercept $= (0, c)$

* No. of times graph cuts x-axis = No. of roots of quadratic

(2 Roots) (1 Root)

* If $a>0$

$(D>0)$ $(D=0)$ $(D<0)$

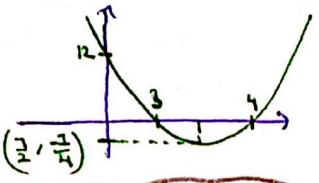
* If $a<0$

$(D>0)$ $(D=0)$ $(D<0)$

Eg: Draw graph of $y = x^2 - 7x + 12$

Sol'n
① $a>0$
② $V = \left(\frac{7}{2}, \frac{-1}{4}\right)$
③ $D = \sqrt{(7)^2 - 4(1)(12)} = 1 > 0 \rightarrow$ 2 Roots
④ y-intercept $= 12$

Graph ⇒



Also, Range of Quad. $= \left[\frac{-1}{4}, \infty\right)$

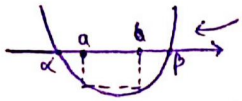
NOTE :

1) If $a>0$, $D<0$ ⇒ $ax^2+bx+c=0$ is always +ve

2) If $a<0$, $D<0$ ⇒ $ax^2+bx+c=0$ is always −ve

Ques If $b>a$, then $(x-a)(x-b)-1=0$ has
(A) Both roots in (a,b)
(B) Both roots in $(-\infty, a)$
(C) Both roots in (b, ∞)
(D) One root in $(-\infty, a)$ and other in (b, ∞).

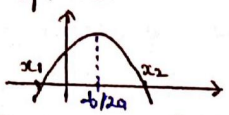
Sol'n $f(x) = (x-a)(x-b)-1$ → Coeff. of $x^2 > 0$
Also, $(f(a) = -1$ & $f(b) = -1)$ ⇒ Upward parabola

⇒ 

← Only possible graph

i.e. α lie in $(-\infty, a)$
β lie in (b, ∞)
} (D) Ans

Ques Consider following graph of $ax^2+bx+c=0$ & select correct options.



(A) $a-b-c<0$
(B) $bc<0$
(C) $b>0$
(D) 'b' & 'c' have same sign but different from 'a'.

Sol'n
⊙ $a<0$
⊙ $\frac{-b}{2a} = +ve \Rightarrow -b = -ve \Rightarrow b = +ve$
⊙ $c>0$

Hence, (A), (C) & (D) are correct options

Symmetric Expressions :

↳ Expressions which do not change upon interchanging α and β.

Example :

1) $\alpha^2+\beta^2$
2) $\alpha^2+\alpha\beta+\beta^2$
3) $\frac{1}{\alpha}+\frac{1}{\beta}$
4) $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$
5) $\alpha^2\beta+\beta^2\alpha$
6) $\alpha^3+\beta^3$
7) $\left(\frac{\alpha}{\beta}\right)^2+\left(\frac{\beta}{\alpha}\right)^2$
8) $\alpha^4+\beta^4$

* These expressions are generally used to form new quadratic with diff. roots

Eg: Let a quadratic be $x^2+2x+3=0$ with α & β as roots. Find quad. whose roots are $1/\alpha$ and $1/\beta$.

Sol'n $\frac{1}{\alpha}+\frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{-2}{3}$; $\frac{1}{\alpha}\cdot\frac{1}{\beta} = \frac{1}{3}$

So, Quad. : $\boxed{x^2 + \frac{2}{3}x + \frac{1}{3} = 0}$ Ans

Shortcut : If roots of 'to be' formed quadratic are symmetric, simply put
x = (New roots)
↳ Get value of α or β & put in given quadratic.

Eg: α & β are roots of $4x^2 - 5x + 7 = 0$. Find quad. whose roots are $(\alpha+2)$ & $(\beta+2)$.

Sol'n $x = \alpha+2 \Rightarrow \boxed{\alpha = x-2}$
$4x^2 - 5x + 7 = 0$ ← Put here as α is 1st root.
$4(x-2)^2 - 5(x-2) + 7 = 0$
$4x^2 + 16 - 16x - 5x + 10 + 7 = 0$

$\boxed{4x^2 - 21x + 33 = 0}$ Ans

Condition for Common Roots :

1) Condition for both Roots Common :→

$a_1x^2 + b_1x + c_1 = 0$ ← α, β
$a_2x^2 + b_2x + c_2 = 0$ ← α, β

Cond' $\boxed{\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}}$

2) Condition for one root Common :→

$a_1x^2 + b_1x + c_1 = 0$ ← α, β
$a_2x^2 + b_2x + c_2 = 0$ ← α, γ
$(\beta \neq \gamma \neq 0)$

To solve such system of quadratic eqn's

(i) Substitute common root (α) in both given quadratics.

(ii) Eliminate term containing α^2 and we will simply get value of α.

(iii) Apply product of roots to find β & γ.

$\left[\alpha\beta = \frac{c_1}{a_1}\right]$ $\left[\alpha\gamma = \frac{c_2}{a_2}\right]$

NOTE : Given one roots is common but one of the QE also has $D<0$, then both roots will be Common.

Quadratic Equation-2

Q.e $a_1x^2+b_1x+c_1=0$ (roots $\alpha=p+iq$, $\beta=p-iq$)

[roots will be conjugate pairs]

$a_2x^2+b_2x+c_2=0$ (roots $\lambda=p+iq$, $\gamma=p-iq$)

Ques Let $\lambda\neq0$ be an R. If α & β are roots of the equation $x^2-x+2\lambda=0$ & α & γ are roots of eqn $3x^2-10x+27\lambda=0$, then find value of $(\beta\gamma/\lambda)$.

Sol^n

$x^2-x+2\lambda=0$ (α, β)

$3x^2-10x+27\lambda=0$ (α, γ)

So,

$3\alpha^2-\alpha+2\lambda=0$]$\times3$

$3\alpha^2-10\alpha+27\lambda=0$

$-7\alpha+21\lambda=0 \Rightarrow \boxed{\alpha=3\lambda}$

Now,

$\alpha\beta=2\lambda$ $\Rightarrow \boxed{\beta=\frac{2}{3}}$

$\alpha\gamma=\frac{27\lambda}{3}$ $\boxed{\gamma=3}$

So, $\frac{\beta\gamma}{\lambda}=\frac{(2/3)(3)}{1/9}\Rightarrow 18$ Ans

Ques Let $a,b\in R$, $a\neq0$ be such that the eqn $ax^2-2bx+5=0$ has a repeated root α, which is also a root of the eqn, $x^2-2bx-10=0$. If β is the other root of this eqn then find $\alpha^2+\beta^2$.

Sol^n

$ax^2-2bx+5=0$ (α, α) ; $x^2-2bx-10=0$ (α, β)

So, $\alpha+\alpha=\frac{-(-2b)}{a}$ and $\alpha\cdot\alpha=\frac{5}{a}$

$\Rightarrow \boxed{\alpha=b/a}$; $\boxed{\alpha^2=5/a}$

$\downarrow$

$\boxed{b^2=5a}$

Now, $\alpha^2-2b\alpha-10=0 \Rightarrow \frac{b^2}{a^2}-2b\left(\frac{b}{a}\right)-10=0$

$b^2-2b^2a-10a^2=0 \Rightarrow (5a)^2-2b(5a)a-10a^2=0$

$\Rightarrow 5a^2-20a^2=0 \Rightarrow 5a=20a^2$

$a=\frac{1}{4}$ ✓ , 0 (X)

So, $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$

$=(2b)^2-2(-10)\Rightarrow 4b^2+20$

$=4\left(5\times\frac{1}{4}\right)^2+20\Rightarrow 25$ Ans

Ques If $a,b,c\in R$ and eqn $ax^2+bx+c=0$ and $x^2+2x+9=0$ have a common root, find $a:b:c$.

Sol^n $x^2+2x+9=0$ → $D=(4)^2-4(1)(9)=16-36=-20<0$

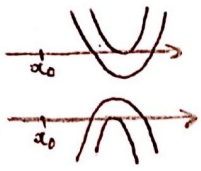
Thus both roots will be equal and

$\frac{a}{1}=\frac{b}{2}=\frac{c}{9}$

$\Rightarrow \boxed{a:b:c=1:2:9}$ Ans

#Location of Roots :

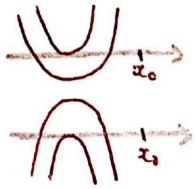
1) Both Roots are greater than x_0 :→



1) $D\geq0$
2) $-b/2a>x_0$
3) $a\cdot f(x_0)>0$

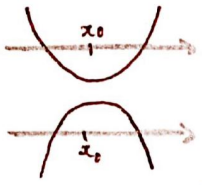
Take intersection of all 3 cond^n.

2) Both Roots less than x_0 :→



1) $D\geq0$
2) $-b/2a<x_0$
3) $a\cdot f(x_0)>0$

3) One root less than x_0, other greater than x_0 :→

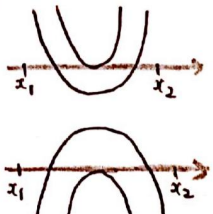


* $D>0$ always as 2 roots must exist for this case
* Can't comment about x-coordinate of vertex

$\Rightarrow \boxed{a\cdot f(x_0)<0}$

Only condition to check when x_0 lie b/w both the roots of quadratic

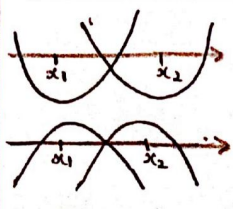
4) Both roots lie b/w x_1 and x_2 :→



1) $D\geq0$,
2) $\boxed{x_1<\frac{-b}{2a}<x_2}$
3) $\boxed{a\cdot f(x_1)>0}$; $\boxed{a\cdot f(x_2)>0}$

Graph can also pass through any one root at a time.

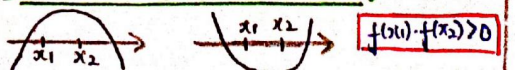
5) Exactly one root b/w x_1 & x_2 :→



$\boxed{f(x_1)\cdot f(x_2)\leq0}$

Equality came bcz graph can also pass through any one root at a time (of x_1 or x_2)

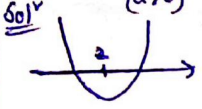
6) Both roots outside x_1 & x_2 :→



$\boxed{f(x_1)\cdot f(x_2)>0}$

Ques Find the value of K for which the one root of eqn $x^2-(K+1)x+K^2+K-8=0$ exceed 2 and other is smaller than 2.

Sol^n

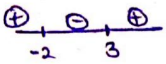


$(a>0)$

$1\cdot f(2)<0$

$1\cdot\left(4-(K+1)2+K^2+K-8\right)<0$

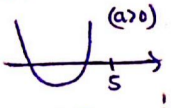
$\Rightarrow K^2-K-6<0 \Rightarrow (K-3)(K+2)<0$



$\boxed{K\in(-2,3)}$ Ans

Ques If both the roots of quadratic eqn $x^2-2Kx+K^2+K-5=0$ are less than 5, then find interval in which K lies.

Sol^n



$(a>0)$

(I) $D\geq0$

$4K^2-4(K^2+K-5)\geq0$

$-4K+20\geq0$

$\Rightarrow \boxed{K\leq5}$ —①

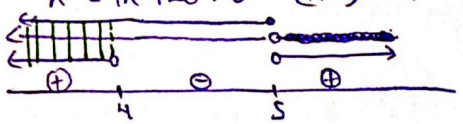
(II) $-b/2a<5$

$\frac{2K}{2}<5$

$\Rightarrow \boxed{K<5}$ —②

(III) $a\cdot f(5)>0 \Rightarrow 25-10K+K^2+K-5>0$

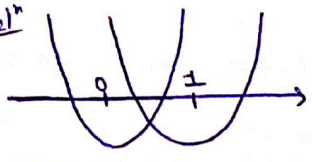
$K^2-9K+20>0 \Rightarrow (K-5)(K-4)>0$



$\Rightarrow \boxed{K\in(4,\infty)}$ Ans

Ques The set of all real values of λ for which the quadratic eqn $(\lambda^2+1)x^2-4\lambda x+2=0$ always have exactly one root in the interval $(0,1)$ is _____.

Sol^n



$f(0)\cdot f(1)\leq0$

$2\left[\lambda^2+1-4\lambda+2\right]\leq0$

$\lambda^2-4\lambda+3\leq0$

$(\lambda-1)(\lambda-3)\leq0 \Rightarrow \lambda\in[1,3]$

But For $\lambda=1$

$2x^2-4x+2=0$

$x^2-2x+1=0 \Rightarrow \boxed{x=1}$

Quad. only has 1 root hence it will have to be removed

$\Rightarrow \boxed{K\in(1,3]}$ Ans

Theory of Equations :

* Alternatively, change the sign of terms (starting from 2nd term) after dividing entire eqn by coeff. of highest degree variable

(I) For Quadratic Eqn :→

$ax^2 + bx + c = 0$

↳ $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$ (−) (+)

* $\alpha + \beta = -b/a$
* $\alpha\beta = c/a$

(II) For Cubic Eqn :→

$ax^3 + bx^2 + cx + d = 0$

↳ $x^3 + \frac{b}{a}x^2 + \frac{c}{a}x + \frac{d}{a} = 0$ (−) (+) (−)

* $\alpha + \beta + \gamma = -b/a$
* $\alpha\beta + \beta\gamma + \alpha\gamma = c/a$
* $\alpha\beta\gamma = -d/a$

(III) Bi-Quadratic Eqn :→

$ax^4 + bx^3 + cx^2 + dx + e = 0$

↳ $x^4 + \frac{b}{a}x^3 + \frac{c}{a}x^2 + \frac{d}{a}x + \frac{e}{a} = 0$ (−) (+) (−) (+)

* $\alpha + \beta + \gamma + \delta = -b/a$
* $\alpha\beta + \beta\gamma + \gamma\alpha + \alpha\delta + \beta\delta + \gamma\delta = c/a$
* $\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -d/a$
* $\alpha\beta\gamma\delta = e/a$

Ques For the cubic equation $x^3 - px^2 + qx - r = 0$, Find
(i) sum of the squares of its roots
(ii) sum of the cubes of its roots

Sol^n $x^3 - px^2 + qx - r = 0$ ⟨ α, β, γ

① $\alpha + \beta + \gamma = p$
② $\alpha\beta + \beta\gamma + \alpha\gamma = +q$
③ $\alpha\beta\gamma = +r$

(i) $= \alpha^2 + \beta^2 + \gamma^2$
$= (\alpha+\beta+\gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$
$= \boxed{p^2 - 2q}$ Ans

(ii) $\because \alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma = (\alpha+\beta+\gamma)(\alpha^2+\beta^2+\gamma^2 - \alpha\beta - \beta\gamma - \alpha\gamma)$

$\Rightarrow \alpha^3 + \beta^3 + \gamma^3 = (p^2 - 2q - q)(p) - 3(r)$

$\alpha^3 + \beta^3 + \gamma^3 = p(p^2 - 3q) - 3r$

$\boxed{\alpha^3 + \beta^3 + \gamma^3 = p^3 - 3pq - 3r}$ Ans